

SHIV VYAS

shivvyas3030@gmail.com | (609) 455-2183 | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

PROFESSIONAL EXPERIENCE

Johnson & Johnson | Consultant (Tata Consultancy Services)

Aug 2022 – Present

Data Science & Machine Learning

- Developed AI-based defect detection models achieving 95% accuracy for contact lens inspection using CNN, deep learning, image analytics, analyzing defect rates & patterns, replacing rule-based systems with scalable ML solutions.
- Built LSTM-based predictive maintenance models using time-series vibration sensor data with Fourier transformation for anomaly detection and fault prediction, reducing machine downtime by 40%.
- Performed EDA on manufacturing data (yield rates, defect rates), engineered features from sensor data, applied clustering, regression, classification & neural network algorithms for process optimization & quality control.

IoT & Data Engineering

- Led architecture of GxP-compliant IoT platform connecting 70+ catheter calibration systems with real-time edge/cloud data pipelines, reducing operational time from 2 days to real-time for J&J and partner sites.
- Managed sensor data collection from edge devices and built complete data acquisition pipelines for image collection, contextualization, and preprocessing within existing manufacturing infrastructure.
- Drove data gap analysis in Databricks processing Big Data (18M+ records) from SQL databases, Azure Delta Lake, and Blob Storage in 24/7 sensor stream environments.

Visualization & Collaboration

- Created data visualizations using Streamlit, Plotly, and PowerBI to communicate findings and collaborated with engineers, surgeons, and domain experts to translate business requirements into data-driven solutions.
- Utilized JIRA for project management following Agile and Scrum methodologies.

Quality Assurance & Testing

- Served as QA Lead for digital surgery platform (100,000+ LoC), leading test framework development & automation for REST APIs, CI/CD (Jenkins), Azure IoT components with regression, integration, and load testing.

Engage AI & Florida Institute of Technology | Data Engineer

Jun 2021 – Feb 2022

- Scraped 1.5M+ socioeconomic data points from UN/IMF/World Bank, performed feature extraction and dataset creation. Built Flask/Python platform with SQL database for analysis and visualization.

Renesas Electronics Americas | Analyst Intern

May 2020 – Aug 2020

- Facilitated semiconductor company integration into Renesas eco-system via gap analysis, testing, and process optimization. Consolidated Oracle/SAP/NetSuite data, reducing order processing time 40%.

Scudo Systems | Co-Founder

Jan 2017 – Jul 2019

- Designed IoT-enabled vehicle security system; won 2017 Start-Up India Yatra among 2,200+ start-ups.
- Raised INR 1.2M in funding for commercialization and production.

TECHNICAL SKILLS

Programming & ML: Python (sklearn, xgboost, pycaret, pandas, numpy), SQL, C++, Java | Neural Networks, Decision Trees, Clustering, Regression, Classification, CNN, RNN, LSTM, NLP | PyTorch, TensorFlow, Keras, OpenCV

Web & APIs: Flask, RESTful APIs, Microservices, HTML, CSS, JavaScript, Websockets

Data & Analytics: Time-Series Analysis, Feature Engineering, Image Analytics, Anomaly Detection, Big Data | PySpark, Databricks, Delta Lake | MySQL, Oracle, PostgreSQL, MongoDB | Streamlit, Tableau, PowerBI

Cloud & Misc: Azure (IoT Hub, Event Hub), Docker, Kubernetes, Ansible, Jenkins, Git | Linux OS, Raspberry Pi

EDUCATION

Master of Science in Computer Science | Florida Institute of Technology

GPA: 3.7 / May 2021

- Teaching Assistant: Taught Python programming and Linux to 100+ sophomores.

Bachelor of Engineering in Computer Engineering | Gujarat Technological University

GPA: 3.6 / Jul 2019